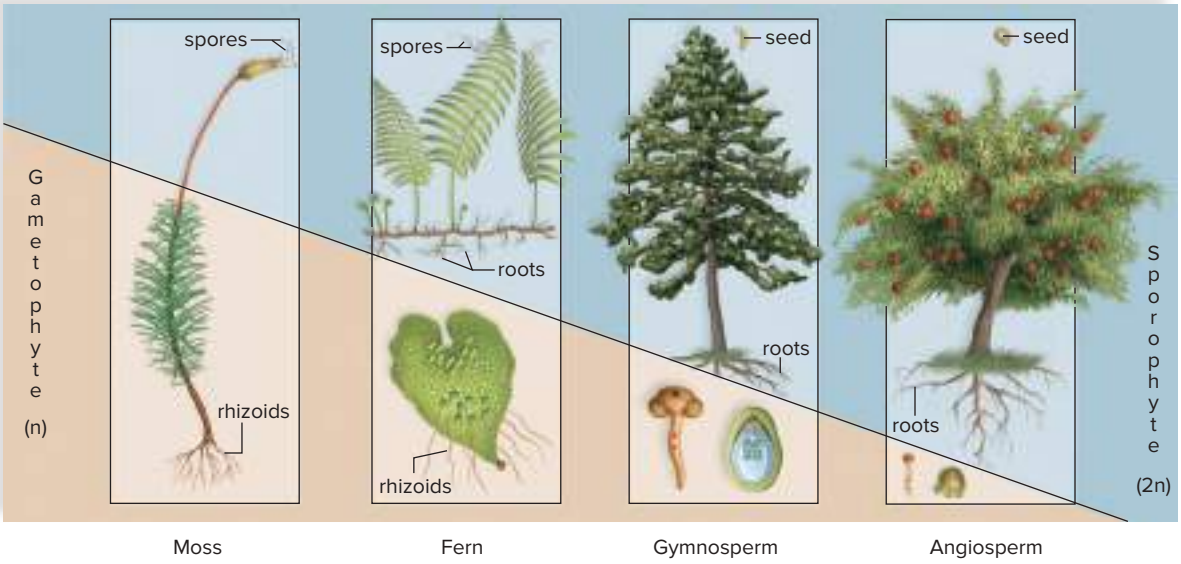


Figure 23.6
Reduction in the size of the gametophyte. Notice the reduction in the size of the gametophyte and the increase in the size of the sporophyte among these representatives of today's land plants. This trend occurred as these plants became adapted for life on land. In the moss and fern, spores disperse the gametophyte. In gymnosperms and angiosperms, seeds disperse the sporophyte.

Check Your Progress 23.1

1. Compare and contrast the traits of charophytes and land plants.
2. List the adaptations that led to a land existence for plants.
3. Identify the role of each generation in the alternation-of-generations life cycle.

23.2 Evolution of Bryophytes: Colonization of Land

Learning Outcomes

Upon completion of this section, you should be able to

1. List the traits that classify a plant as a bryophyte.
2. Compare the three groups of bryophytes.
3. Identify the key structures and stages in the life cycle of a moss.

The **bryophytes**—liverworts, hornworts (Anglo-Saxon *wort,* "herb"), and mosses—were the first plants to colonize land. They only superficially appear to have roots, stems, and leaves, because, by definition, true roots, stems, and leaves must contain vascular tissue. **Vascular tissue** is specialized for the transport of water and organic nutrients throughout the body of a plant. Bryophytes, which lack vascular tissue, are often called **nonvascular plants.** Vascular tissue also provides support to the plant body, so bryophytes typically are low-lying; some mosses reach a maximum height of only about 20 cm.

The fossil record contains some evidence that the various bryophytes evolved during the Ordovician period (about 443.7 MYA, see Table 18.1). An incomplete fossil record makes it difficult to tell how closely related the various bryophytes are. Molecular data, in particular, suggest these plants have individual lines of descent, as shown in Figure 23.1, and that they do not form a monophyletic group. The observation that today's mosses have a rudimentary form of vascular tissue suggests they are more closely related to vascular plants than to hornworts and liverworts.

Bryophyte bodies are covered by a cuticle, which is interrupted in hornworts and mosses by stomata, and they have apical tissue that produces complex tissues. However, bryophytes are the only land plants in which the gametophyte is dominant (see Fig. 23.6). Antheridia produce flagellated sperm, which means they need a film of moisture in order to swim to eggs located inside archegonia. The bryophytes' lack of vascular tissue and the presence of flagellated sperm mean you are apt to find bryophytes in moist locations. Some bryophytes compete well in harsh environments because they can reproduce asexually.

Bryophytes contribute to the lush beauty of forests. They can convert mountain rocks to soil, hold moisture and metals, and resist desiccation. Scientists working in genetic engineering are interested in bryophytes' ability to resist chemical reagents, decay, and herbivory. The idea is that these traits could be transferred to other plants.

Liverworts

Liverworts are divided into two groups—the thallose liverworts with flattened bodies, known as a thallus; and the leafy liverworts, which superficially resemble mosses. The name *liverwort* refers to the lobes of the thallus, which to some resemble the lobes of the liver. The majority of liverwort species are the leafy types.

The liverworts in the genus *Marchantia* have a thin thallus, about 30 cells thick in the center. Each branched lobe of the thallus is approximately a centimeter in length; the upper surface is divided into diamond-shaped segments with a small pore, and the lower surface bears numerous hairlike extensions called **rhizoids** (Gk. *rhizion,* dim. of "root") that project into the soil (Fig. 23.7). Rhizoids serve as anchors and assist in the absorption of water.

Marchantia species reproduce both asexually and sexually. Gemmae cups on the upper surface of the thallus contain gemmae,

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a. Thallus with gemmae cups

b. Male gametophytes bear antheridia

c. Female gametophytes bear archegonia

Figure 23.7 Liverwort, *Marchantia*. **a.** Gemmae can detach and start a new plant. **b.** Antheridia are present in disk-shaped structures. **c.** Archegonia are present in umbrella-shaped structures. (a, c): Ed Reschke/Stone/Getty Images; (b): J. M. Conrader/Science Source

groups of cells that detach from the thallus and can start a new plant. Sexual reproduction depends on disk-headed stalks that bear antheridia, as well as on umbrella-headed stalks that bear archegonia. Following fertilization, tiny sporophytes, composed of a foot, a short stalk, and a capsule, begin growing within archegonia. Wind-blown spores are produced within the capsule.

Hornworts

The **hornwort** gametophyte usually grows as a thin rosette or ribbon-like thallus between 1 and 5 cm in diameter. Although some species of hornworts live on trees, most live in moist, well-shaded areas. They photosynthesize, but also have a symbiotic relationship with cyanobacteria, which, unlike plants, can fix nitrogen from the air.

The small sporophytes of a hornwort resemble tiny green horns rising from a thin gametophyte, usually less than 2 cm in diameter (Fig. 23.8). Like the gametophyte, the sporophyte can photosynthesize, although it has only one chloroplast per cell. Meiosis occurs within the "horn," and spores are released. Alternatively, hornworts can bypass alternation of generations by reproducing asexually through fragmentation.

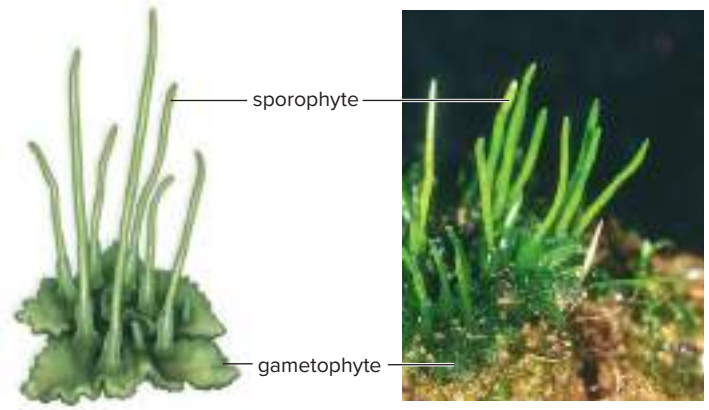


Figure 23.8 Hornwort, *Anthoceros* sp. The "horns" of a hornwort are photosynthetic sporophytes that grow continuously from a base anchored in gametophyte tissue. (photo): Steven P. Lynch

Mosses

Mosses are the largest group of nonvascular plants, with over 12,000 species. There are three distinct groups of mosses: peat mosses, granite mosses, and true mosses. Although most prefer damp, shaded locations in the temperate zone, some survive in deserts, and others inhabit bogs and streams. In forests, they frequently form a mat that covers the ground and rotting logs. In dry environments, they may become shriveled, turn brown, and look completely dead. As soon as it rains, however, the plant becomes green and resumes metabolic activity.

Members of the moss genus *Sphagnum* (peat moss) have great commercial and ecological importance to humans. The cell walls of peat moss have a tremendous ability to absorb water, which is why it is used in gardening to improve the water-holding capacity of the soil, and why it was used historically as wound dressing on battlefields. One percent of Earth's surface is peatlands, where dead *Sphagnum* accumulates but does not decay.

Figure 23.9 describes the life cycle of a typical temperate-zone moss. The gametophyte of mosses begins as an algalike, branching filament of cells, the protonema, which precedes and produces upright, leafy shoots that sprout rhizoids. The shoots bear either antheridia or archegonia. The dependent sporophyte consists of a foot, which is enclosed in female gametophyte tissue; a stalk; and an upper capsule, which contains the sporangium, where spores are produced. A moss sporophyte is always attached to the gametophyte. At first, the sporophyte is green and photosynthetic; at maturity, it is brown and nonphotosynthetic. In some species, the sporangium can produce as many as 50 million spores. The spores disperse the new gametophyte generation.

Check Your Progress 23.2

1. Explain the various methods of bryophyte reproduction.
2. List the characteristics that enabled the bryophytes to successfully colonize land.
3. Explain which portion of the bryophyte life cycle is the most dominant.

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